

ALC - MIARFID

Laboratory Lesson 3

Sexism Identification in TikTok Videos

12th March 2025

Objectives

In this laboratory session, we aim to design and implement a computational system capable of automatically detecting sexism in TikTok videos. Consistent with the scope of the subject, this system will use text-based computational linguistic techniques combined with multimodal signal processing.

More specifically, **we will work with video transcriptions and the sensorial data introduced in previous sessions**. What is more important now is that the current laboratory session takes a notably different perspective:

- **Exploring the training and evaluation** of automatic sexism detection models **in multi- and cross-lingual scenarios**. In real-world applications, data scarcity often makes it challenging to train models specifically tailored to a target language. To familiarize yourself with these situations, this laboratory session proposes the following training configurations:
 - *i) Training with only Spanish Samples.*
 - *ii) Training with only English Samples.*
 - *iii) Training with both Spanish and English Samples.*

By evaluating all three models on the full test set, which includes both Spanish and English samples, we can assess their robustness in cross-lingual evaluation scenarios and multilingual training settings. During this laboratory session, you are encouraged to conduct performance analyses per language to explore model architecture design improvements. **Note, however, that at the time of submission, you must include the predictions for all test samples from each proposed model, as detailed at the end of this bulletin.**

The EXIST2026 Challenge

This lab session is designed to simulate the participation in the **Task 3.1: Sexism Identification in TikTok videos** of the EXIST 2026 Challenge¹. This binary classification task, as detailed below, includes videos in both Spanish and English, encouraging the development of multilingual models. Here you can see an example of the memes we can find in the training dataset of EXIST2026:

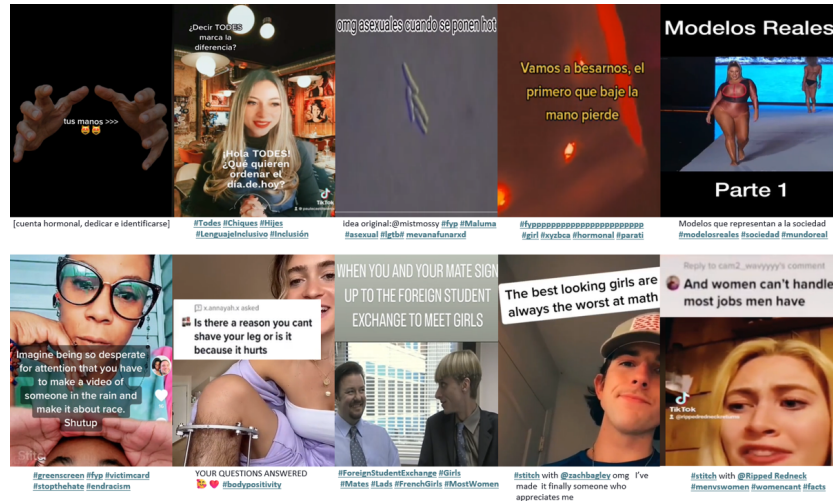


Figure 1: Examples of TikTok Screenshots in the EXIST 2026 dataset, showcasing the inherent challenges of the proposed multimodal and multilingual task.

However, please note that **in this laboratory, we will focus on text-level and sensorial analysis**, using the transcriptions of the audio associated with the videos and their associated eye-tracking, EEG, and heart rate metadata. These transcriptions not only capture the spoken message of the target person in the video, but may also include additional elements, such as sound event detection comments (e.g., keywords for clapping, laughing, or music).

Optionally, as in the previous laboratory work, you may also work with video and/or audio data if you are interested in exploring multi-modal approaches. More details can be found in the next section describing the lab materials. However, please note that this will not impact the evaluation of your laboratory work.

¹<https://nlp.uned.es/exist2026/>

Lab Materials

All necessary data and metadata required for this laboratory session can be accessed through the subject’s PoliformaT platform, where we can find the:

- `dataset_task3_exist2026/` directory, which contains two JSON files specifying our training and test samples of the EXIST2026 Task 3.1 challenge dataset, along with their associated metadata. Among this metadata, apart from the sample ID and multiple labels, we can find the *text* keyword, which contains the Whisper²-based transcription of the audio associated with the video. *Please note, unlike the image captions from the previous session, transcriptions are in the target language of each sample.* Besides, we can also find the *sensorial* field, which stores the multimodal features we described in the previous laboratory session.
- `golds_task3_exist2026/` directory, which contains a JSON file specifying the gold standards for the training set. They are the reference ground-truth labels to evaluate the performance of your models.

An important point to note is that, as in a real challenge, **labels for the test set are not provided**. *Please refer to the implementation details outlined in the first laboratory session* to address strategies for handling model evaluation, multiple annotation labels, and the use of PyEvALL.

If you’re interested, you can get access to the TikTok videos of the EXIST2026 challenge by following the instructions on the official web page: <https://nlp.uned.es/exist2026/>.

Please note that you should filter the data according to the provided data splits (training and test) for this laboratory work.

What Should You Do?

In this lab session, you will: (i) assess the effectiveness of multimodal approaches for sexism detection in videos; (ii) explore the training and evaluation of these methods under cross- and multilingual conditions; and (iii) familiarize yourself with more challenging real-world scenarios. This may involve reading scientific papers related to the task or efficient training and adaptation methods in multilingual contexts to gather insights on potentially useful approaches.

For reference, you can consult the official EXIST2025 challenge overview paper [1]. In particular, focus on Subsection 6.7.2 (Table 19), where you will find the leaderboard with the final ranking of the task under hard-label settings. A brief description of the models involved is found in Section 5. Remember, the aim of this lab session is not to surpass the state of the art, but to learn. Once you finish your experiments, you must submit via PoliFormaT:

²<https://github.com/openai/whisper>

- **A ZIP file with the 3 proposed models.** This compressed file should thereby contain three JSON files, one for each of the proposed models: (i) *Trained with only Spanish samples*; (ii) *Trained with only English samples*; and (iii) *Trained with both Spanish and English samples*. While per-language analysis can be conducted during this laboratory work, **each JSON file must include the model’s predictions for the entire test dataset** — considering both languages — formatted according to the guidelines provided in the first lab bulletin. *This will be part of our in-home challenge :) Who will develop the most robust approach?* Please ensure that your JSON files are well-organized, as exemplified by the image below, from which we can infer that we should follow the naming template: `<GROUP-ID>_<MODEL-ID>_trainedon{both|spanish|english}.json`³.

```
alc_lab3_whatareyoutiktokingabout/
├─ whatareyoutiktokingabout_trainedonboth.json
├─ whatareyoutiktokingabout_trainedonenglish.json
└─ whatareyoutiktokingabout_trainedonspanish.json
```

- **A PDF report (max. 3 pages) briefly describing your approach.** This should include not only details on the methods explored in your work but also your findings, challenges, and insights regarding cross-language settings. Additionally, reflect on why your approach **is** or **is not** working — perhaps analyzing the data distribution might provide valuable clues. Not everything is about training models! :)

Deadline: 1st April 2026, 23.55h

You can work individually or in groups of up to two people.

In the case of group submissions, only one member should upload the final work, clearly indicating the names of all participants in the corresponding task available and prepared in PoliFormat.

References

- [1] Laura Plaza, Jorge Carrillo-de Albornoz, Iván Arcos, Paolo Rosso, Damiano Spina, Enrique Amigó, Julio Gonzalo, and Roser Morante. Overview of EXIST 2025: Learning with Disagreement for Sexism Identification and Characterization in Tweets, Memes, and TikTok Videos. In *International Conference of the Cross-Language Evaluation Forum for European Languages (CLEF)*, pages 266–289. Springer, 2025. [Link to PDF](#).

³**Warning:** any model name that can be considered offensive or inappropriate will result in disqualification and a score of 0 for the laboratory work.